

Non-Local Equilibrium vs Noise: Direct Simulation Monte Carlo of Rarefied Gases

David A. Aji and S. Kokou Dadzie

School of Engineering and Physical Sciences, Heriot-Watt
University, Edinburgh, EH14 4AS Scotland, UK



Journal: Frontiers in Mechanical Engineering, may be published
(2026)

'Noise' in DSMC

Fig 3: Slip velocity magnitude, $K_n = 0.05$, $Ma = 2$.

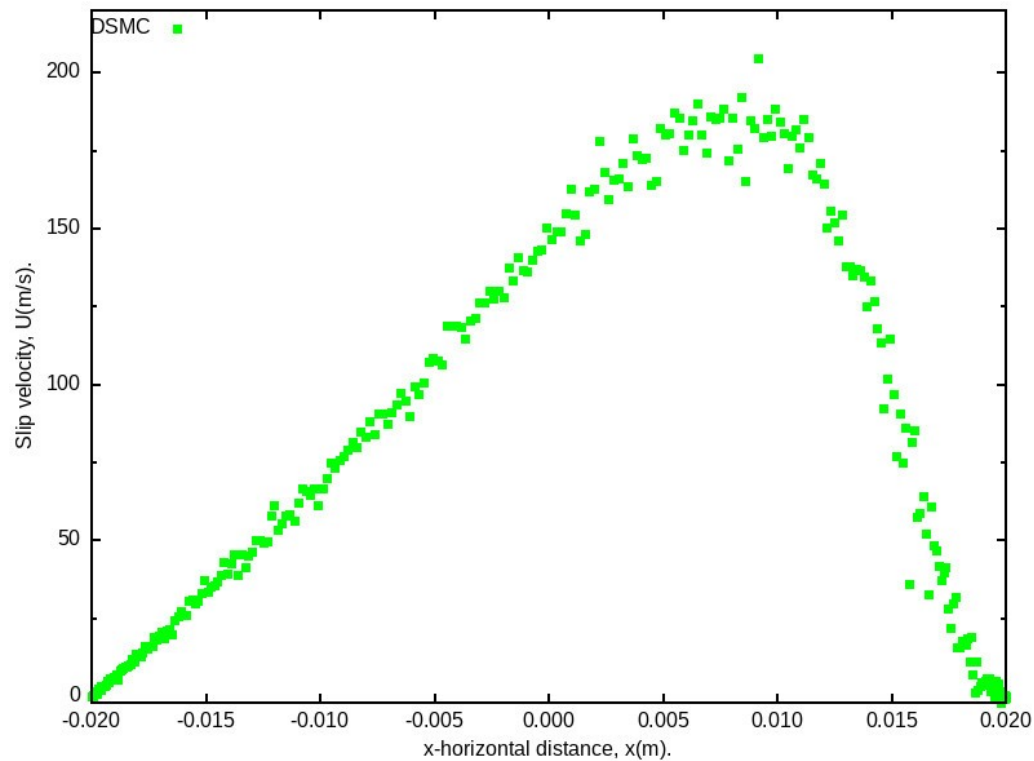
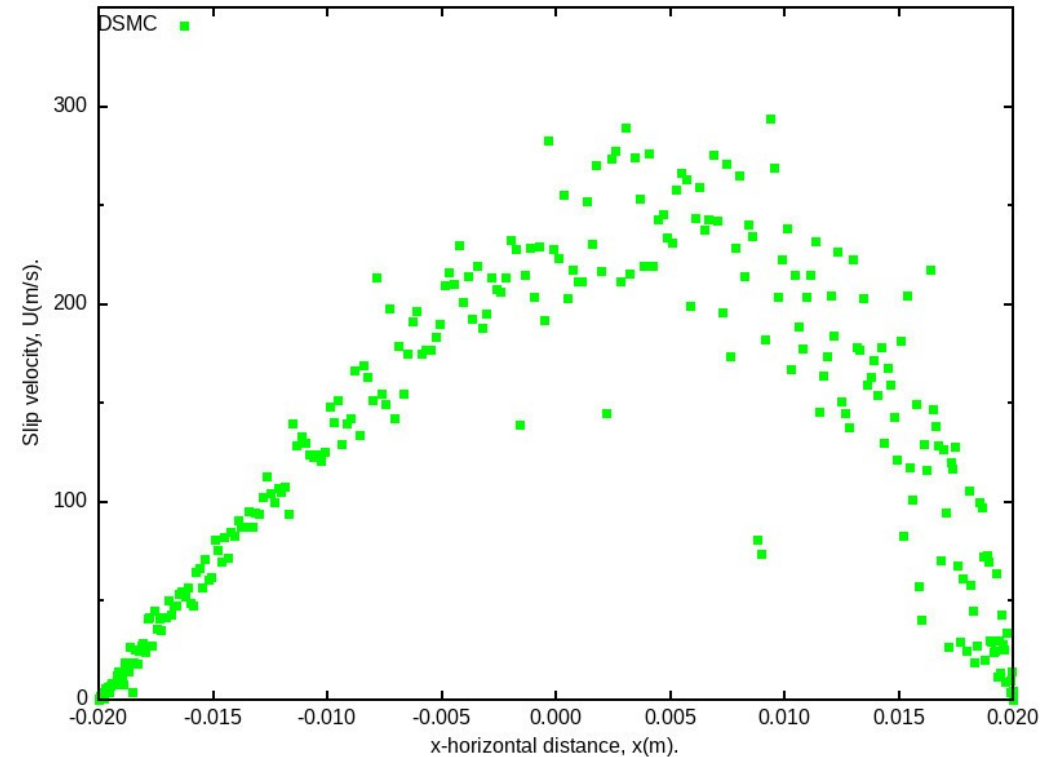


Fig 4: Slip velocity magnitude, $K_n = 2$, $Ma = 2$.



Rarefied gas DSMC noise increases as the flow's Knudsen number rises. Are these normal statistical noises?

‘Noise’ in DSMC: Existing attempts in the literature

- **Kumar et al. Journal of Spacecraft and Rockets (2018), applied a post-processing technique based on POD to denoise DSMC data.**

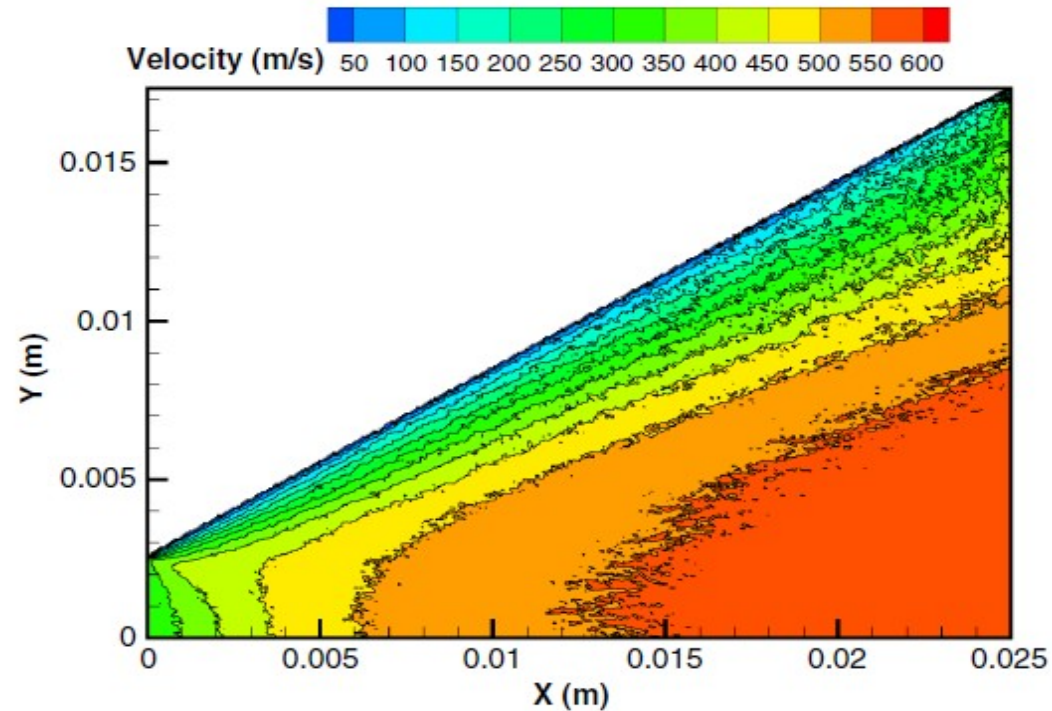
Variables such as velocity, pressure, temperature, and Mach number were compared before and after post-processing, and the algorithm achieved converged, low-scatter results in less computation time.

- **M. Sadr and N. G. Hadjiconstantinou, A variance-reduced direct monte carlo simulation, Journal of Computational Physics 472, 111677 (2023)**

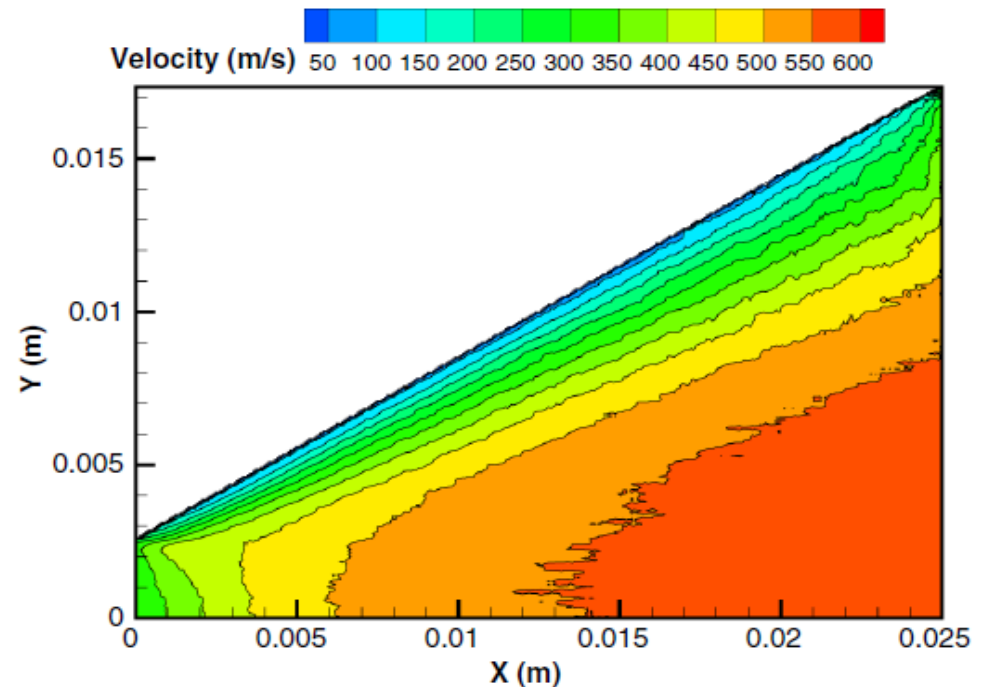
This is a stronger modification to the DSMC method itself.

‘Noise’ in DSMC: Existing attempts in the literature

Fig 7: Kumar et al., Journal of Spacecraft and Rockets, 55(4) 841-847 (2018).



a) DSMC result without postprocessing



b) DSMC result after postprocessing

'Noise' in DSMC: Standard statistical mechanics vs non-local equilibrium

- Fractional error in temperature:

$$E_T = \frac{\sigma_T}{T_0} = \frac{\sqrt{(\delta T)^2}}{T_0 M} = \frac{1}{\sqrt{M N_0}} \sqrt{\frac{k_B}{c_V}}$$

- Transport coefficients:

It appears that these errors in standard statistical mechanics directly define transport coefficients: viscosity and the heat conduction coefficient.

Local Boltzmann-Maxwellian distribution:




Ideal gas:

$$f(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-\frac{v^2 M}{2RT}}$$

Real gas:

$$f(v) = 4\pi \left(\frac{M}{2\pi p V_m} \right)^{3/2} v^2 e^{-\frac{v^2 M}{2p V_m}}$$

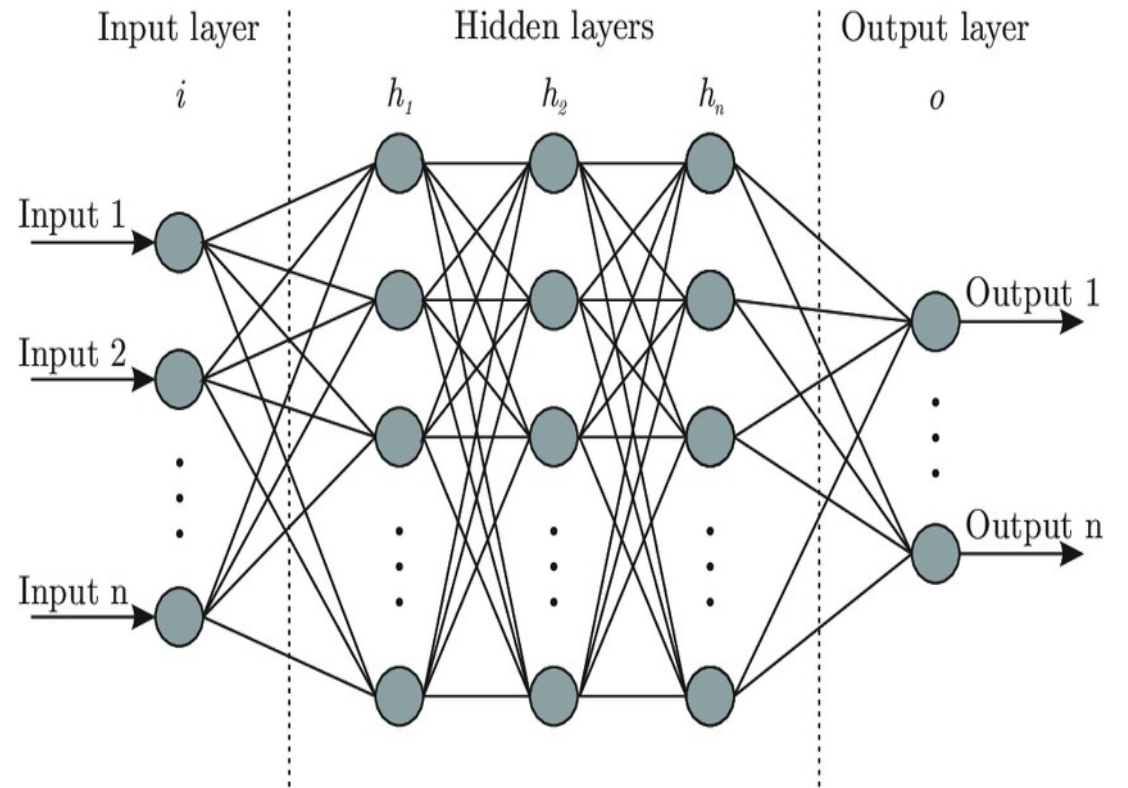
- Defines local thermodynamic properties: $T(x)$ and $p(x)$; $(V_m(x))$

Rarefied gas DSMC noise increasing with Knudsen number rises are non-local equilibrium effect related

Present Methodology

- **ANN** - consists majorly of three layers (input, hidden, and output). The hidden layer houses the major computations.
- Several parameters, such as the learning rate, the number of neurons, the activation function, and so on, that need to be considered when training the model to achieve the desired outcome (output).
- **This training is done with low Knudsen number simulations which is then used to predict high Knudsen number profiles.**

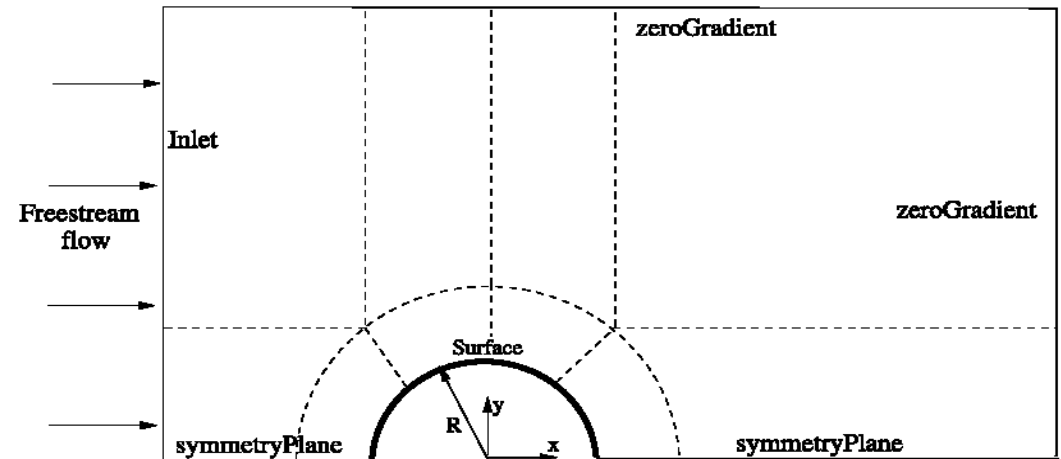
Fig 9: Simple ANN structure



Present Methodology

Fig 10: Geometry of the case studied.

- The case studied is an hypersonic argon gas flow past a circular cylinder in cross-flow.
- Kn_{∞} : 0.05, 0.5, 2. Ma_{∞} : 2 and 6.
- It concerns the prediction of surface quantities such as slip velocity and gas temperature at the surface.



Results: Non-local equilibrium vs noise

Fig 11: Slip velocity magnitude, $K_n = 2$, $Ma = 2$: **Without ANN**

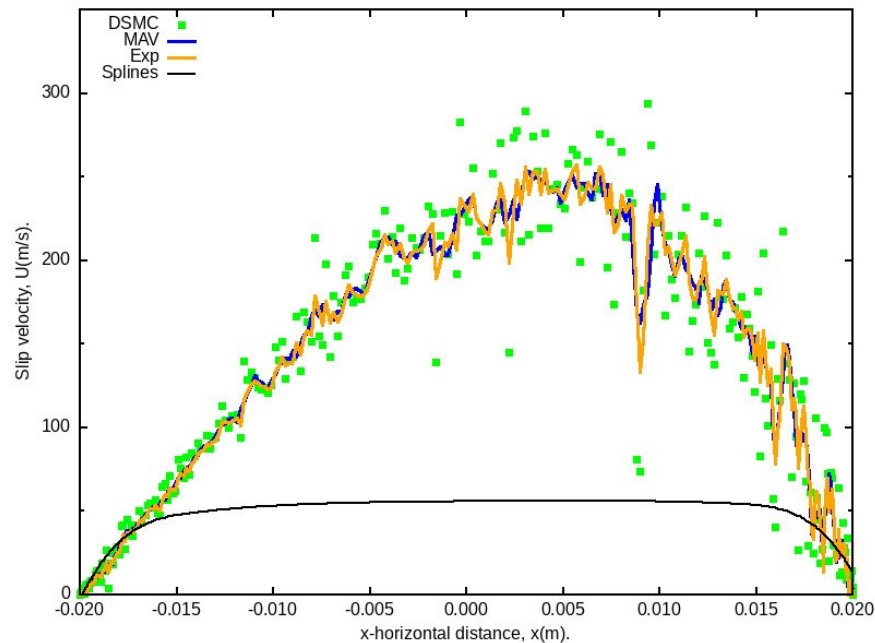
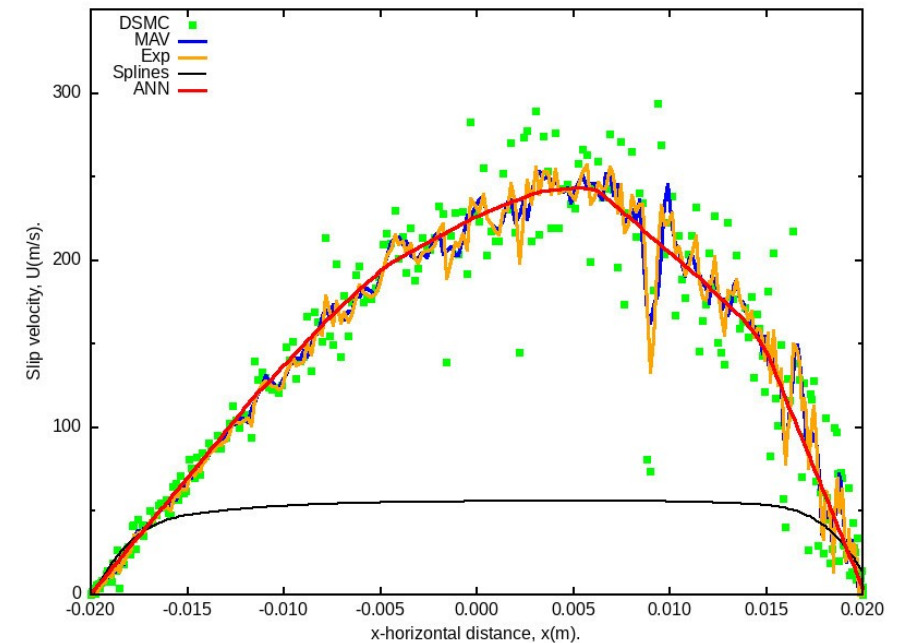


Fig 12: Slip velocity magnitude, $K_n = 2$, $Ma = 2$: **With ANN**



ANN (Machine Learning) extracts actual physical profiles. It is different from standard regression methods

Results: Non-local equilibrium vs noise

Fig 13: Slip velocity magnitude, $K_n = 2$, $Ma = 6$: **Without ANN**

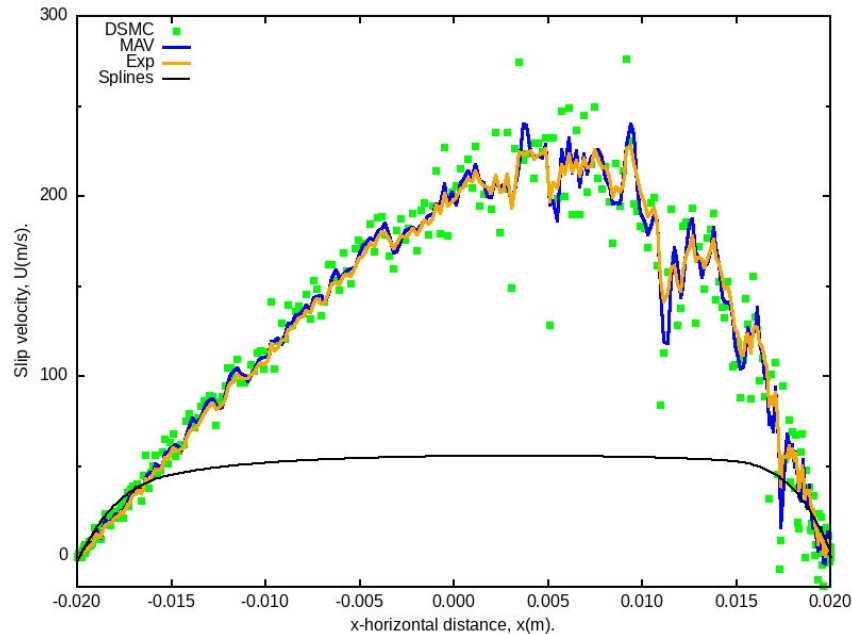
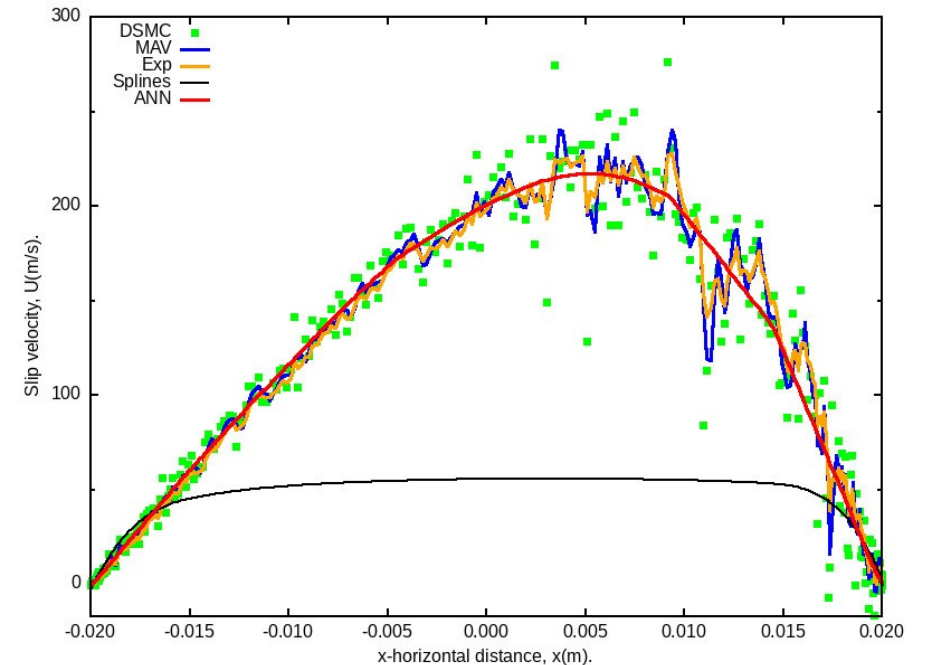


Fig 14: Slip velocity magnitude, $K_n = 2$, $Ma = 6$: **With ANN**



ANN (Machine Learning) extracts actual physical profiles. It is different from standard regression methods

Results: Non-local equilibrium vs noise

Fig 15: Slip velocity magnitude, $K_n = 0.05$, $Ma = 2$: **ANN vs Smooth Bezier**

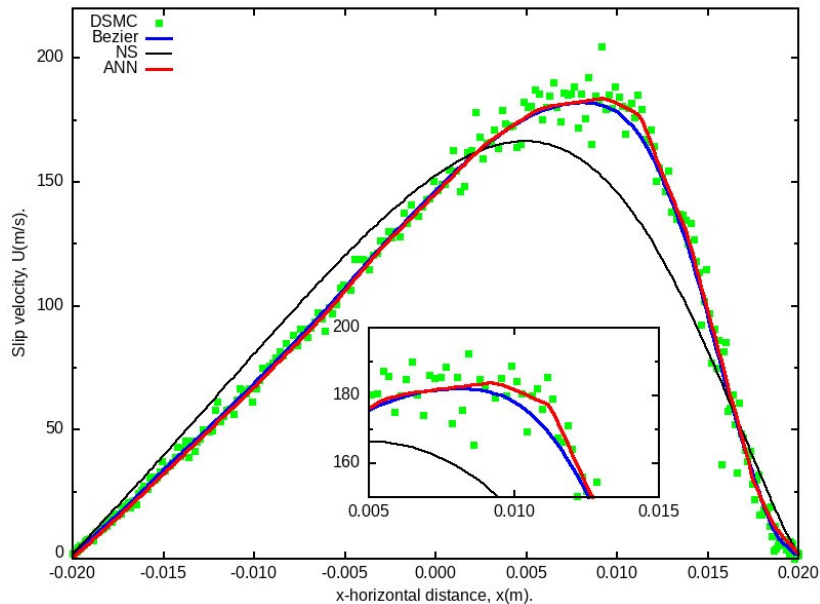
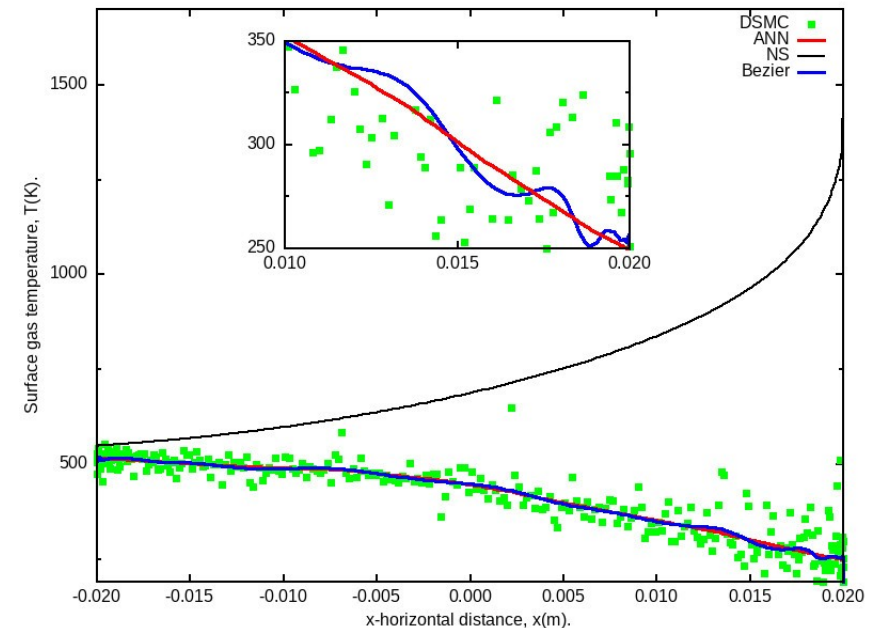


Fig 12: Fig 16: Surface gas temperature, $K_n = 0.05$, $Ma = 2$: **ANN vs Smooth Bezier**



ANN (Machine Learning) extracts actual physical profiles. It is different from a mere data fitting

Results: Non-local equilibrium vs noise

Fig 17: Slip velocity magnitude, $K_n = 2$, $Ma = 6$: **ANN vs Smooth Bezier**

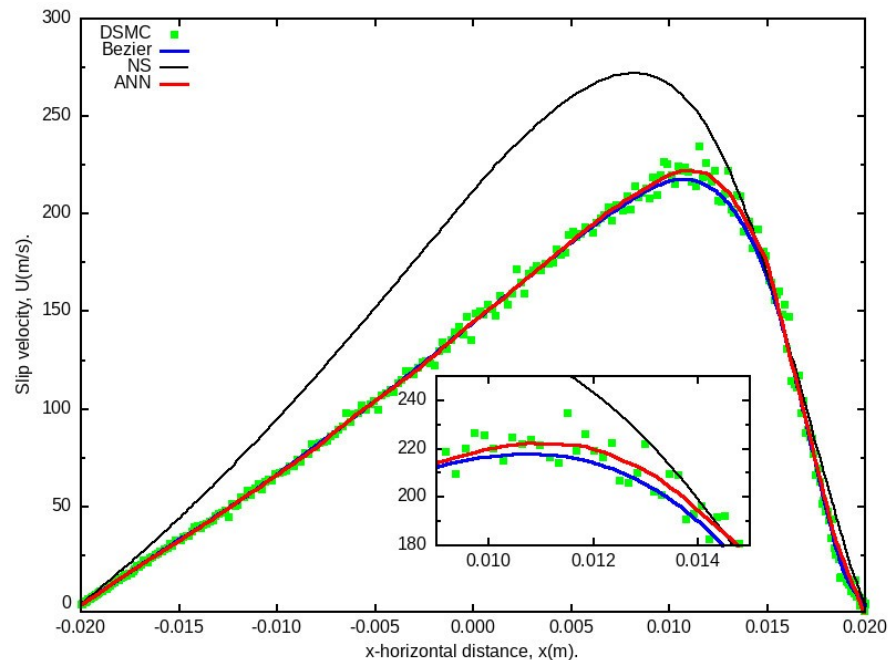
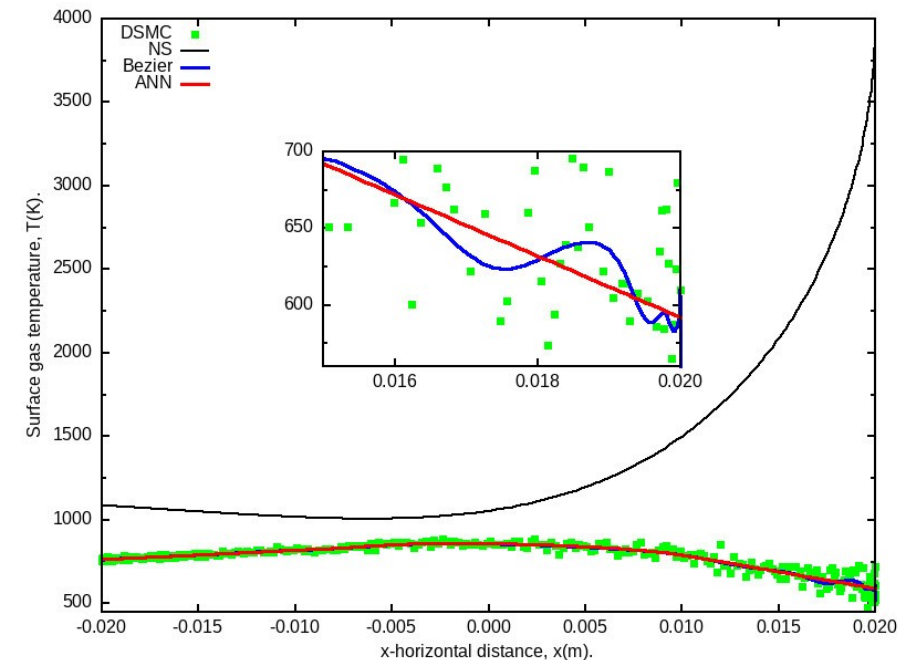


Fig 18: Surface gas temperature, $K_n = 2$, $Ma = 6$: **ANN vs Smooth Bezier**



ANN (Machine Learning) extracts actual physical profiles. It is different from a mere data fitting

Conclusion

- Noise in Direct Simulation Monte Carlo (DSMC) data is correlated with non-local equilibrium in rarefied gas flows.
- We developed an ANN (artificial neural network) algorithm to cope with this noise.
- ANNs may offer appropriate tools to profile high Knudsen number flows at low costs. **Not just fitting the data!**